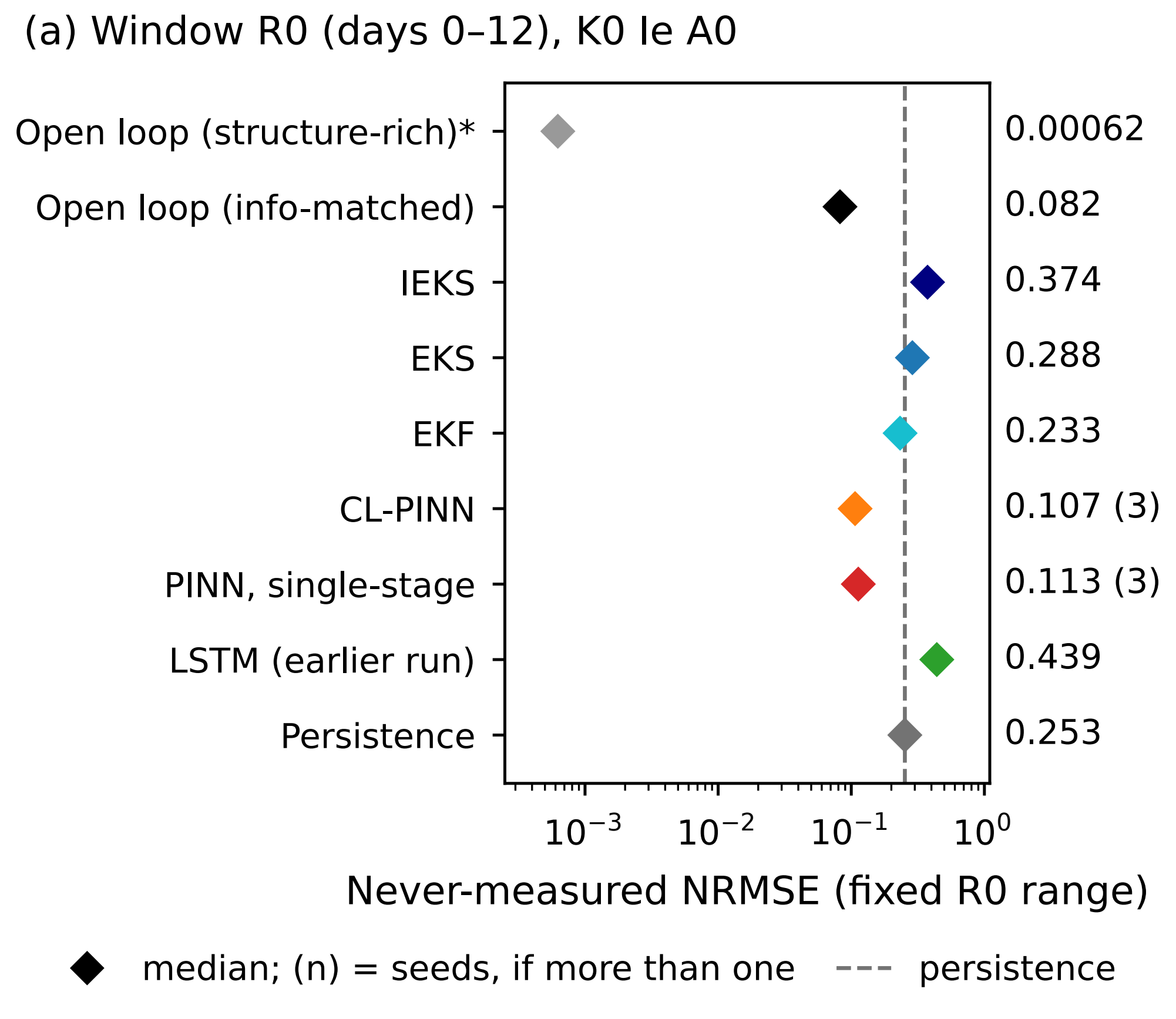
(a) Window R0 (days 0–12), K0 Ie A0
Open loop (structure-rich)*
Open loop (info-matched)
IEKS
EKS
EKF
CL-PINN
PINN, single-stage
LSTM (earlier run)
Persistence
0.00062
0.082
0.374
0.288
0.233
0.107 (3)
0.113 (3)
0.439
0.253
10^{-3}
10^{-2}
10^{-1}
10^{0}
Never-measured NRMSE (fixed R0 range)
median; (n) = seeds, if more than one
persistence